

Eenheid 3.6

Note Title

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Afgeleides van logaritmische funksies

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p218

Ons gaan 3 Stellings bewys.

(Moet kan neerskryf en gebruik)

$$\textcircled{1} \quad \frac{d}{dx}(\ln x) = \frac{1}{x} = \frac{d}{dx}(x)$$

Bewys:

$$y = \ln x \Leftrightarrow e^y = e^{\ln x} = x$$

$$\Leftrightarrow \boxed{e^y = x}$$

$$\frac{d}{dx}(e^y) = \frac{d}{dx}(x) \Rightarrow e^y \times \left(\frac{dy}{dx}\right) = 1$$

(implisiet)

$$\Rightarrow \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x} \quad \therefore \frac{d}{dx}(\ln x) = \frac{1}{x}$$

② $\frac{d}{dx} \ln |x|$

Definise

Bewijs: $y = \ln |x| = \begin{cases} \ln x & \text{as } x > 0 \\ \ln(-x) & \text{as } x < 0 \end{cases}$

• As $x > 0$: $\frac{d}{dx} \ln |x| = \frac{d}{dx} (\ln x) = \frac{1}{x}$
(int stelling!)

• As $x < 0$: $\frac{d}{dx} \ln |x| = \frac{d}{dx} (\ln(-x)) = \frac{\frac{d}{dx}(-x)}{-x} = \frac{-1}{-x} = \frac{1}{x}$

• Dus $\frac{d}{dx} \ln |x| = \frac{1}{x}$

$$\textcircled{3} \frac{d}{dx} \log_a x = \underline{\underline{\frac{1}{x \ln a}}} \quad (\text{Gebruik nie dikwels})$$

Bewys: $\exists^e \log_a x = y \Leftrightarrow \boxed{a^y = x} \quad (\text{definisie})$

$$\frac{d}{dx} (a^y) = \frac{d}{dx} (x) \Rightarrow a^y \ln a \frac{dy}{dx} = 1$$

implisiet

$$\Rightarrow \frac{dy}{dx} = \frac{1}{a^y \ln a}$$

$$= \frac{1}{x \ln a} \quad \parallel$$

VOORBEELDE:

Antoon: kettingreël • Notatie

$$\textcircled{1} \frac{d}{dx} \ln(\sin x^3) = \frac{1}{\sin x^3} \times \cos x^3 \times 3x^2$$

• $y = \ln(\sin x^3) \Rightarrow y' =$

$$\textcircled{2} \frac{d}{dx} \sin(\ln x^3) = \cos(\ln x^3) \times \frac{3x^2}{x^3} \quad \left(= \cos(\ln x^3) \times \frac{1}{x^3} \times 3x^2 \right) (x^3)$$

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$$\frac{d}{dx} \sin(\ln x^3) = \frac{d}{dx} \sin(3 \ln x) = \cos(3 \ln x) \times 3 \times \frac{1}{x}$$

$$\textcircled{3} \frac{d}{dx} (\ln 6^x) = \frac{6^x \ln 6}{6^x} \quad \text{of} \quad \frac{d}{dx} (x \ln 6) = \ln 6$$

getal

④ $\frac{d}{dt} \left[\frac{\ln(1-3t)}{1-\ln 3t} \right]$ (Kwosientreeël)

$$= \frac{\left[\frac{-3}{1-3t} \right] \times (1-\ln 3t) - \ln(1-3t) \times \left[0 - \frac{3}{3t} \right]}{(1-\ln 3t)^2}$$

⑤ $\frac{d}{dt} [t^4 \times \ln(1-e^{4t})]$ (Produkreeël)

$$= (4t^3) (\ln(1-e^{4t})) + (t^4) \left(\frac{1}{1-e^{4t}} \times (-e^{4t} \times 4) \right)$$

OPΣOMMING:

$$\textcircled{1} \cdot \frac{d}{dx} [f(x)]^n = n[f(x)]^{n-1} \times f'(x) \quad \text{kettingreël}$$

Bv.

$$\text{i)} \quad \frac{d}{dx} (1 - \tan 4x)^5 = 5(1 - \tan 4x)^4 \times -\sec^2 4x \times 4$$

$$\text{ii)} \quad \frac{d}{dt} \sin^{\textcircled{4}} 2t = \frac{d}{dt} (\sin 2t)^4 = 4(\sin 2t)^3 \times \cos 2t \times 2$$

$$\text{iii)} \quad \frac{d}{dx} \sqrt[5]{\ln x} = \frac{d}{dx} (\ln x)^{1/5} = \frac{1}{5} (\ln x)^{1/5 - 5/5} \times \frac{1}{x} \\ = \frac{1}{5} (\ln x)^{-4/5} \times \frac{1}{x}$$

$$\text{iv)} \quad \frac{d}{dx} \cos^{\textcircled{2}}(\ln 3x) = 2 \cos(\ln 3x) \times -\sin(\ln 3x) \times \frac{3}{3x}$$

② $\frac{d}{dx} a^{\underline{f(x)}} = \underline{a^{\underline{f(x)}}} \ln \underline{a} \times \underline{f'(x)}$, a70
Formule

B.v. (i) $\frac{d}{dx} 4^{\sin 3x} = 4^{\sin 3x} \ln 4 \times \cos 3x \times 3$

(ii) $\frac{d}{dx} 5^{\cot 3x} = 5^{\cot 3x} \ln 5 \times -\operatorname{cosec}^2 3x \times 3$

(iii) $\frac{d}{dt} 3^{\ln t^4} = 3^{\ln t^4} \ln 3 \times \frac{4t^3}{t^4}$

(iv) $\frac{d}{dx} 3^{\ln(\cos x)} = 3^{\ln(\cos x)} \ln 3 \times \left(\frac{-\sin x}{\cos x} \right) = -\tan x$

$$\textcircled{3} \frac{d}{dx} [f(x)]^{g(x)} \quad ???$$

Geen Formule ???

~~Plan~~ Gebruik logaritmiëse differentiatie

Bv. (i) $y = (4x)^{\cos 3x}$

$$\ln y = \ln (4x)^{\cos 3x}$$

$$\Rightarrow \textcircled{\ln y}_{\text{implicit}} = \cos 3x \times \ln 4x$$

(produkt)

$$\Rightarrow \frac{1}{y} \times \textcircled{\frac{dy}{dx}} = -3 \sin(3x) \times \ln 4x + \cos 3x \times \frac{4}{4x}$$

$$\Rightarrow \frac{dy}{dx} = \underbrace{\left(\frac{y}{y} \right)}_{\checkmark} \left[-3 \sin(3x) \ln 4x + \frac{\cos 3x}{x} \right]$$

(ii) $\frac{d}{dx} (\cos x)^{x^3}$

Sei $y = (\cos x)^{x^3}$

Dan $\ln y = \ln (\cos x)^{x^3}$

$$\Rightarrow \frac{d}{dx}(\ln y) = \frac{d}{dx}(x^3 \ln \cos x) \quad (\text{implicit diff})$$

produkt

$$\Rightarrow \frac{1}{y} \times \frac{dy}{dx} = 3x^2 \ln \cos x + x^3 \times \frac{-\sin x}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} = y [3x^2 \ln \cos x - x^3 \tan x]$$

Gebruik logaritmische differentiatie
om $\frac{dy}{dx}$ te bepaal als

① $y = \frac{6^x \sin 2x}{e^{9x} \tan 4x}$ (komplekse kwosient met produkt. Geen + of -)

$$\ln y = \ln [\text{reëls}]$$

$$\Rightarrow \ln y = \ln 6^x + \ln \sin 2x - \ln e^{9x} - \ln \tan 4x$$

(impliete diff)

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{6^x \ln 6}{6^x} + \frac{2 \cos 2x}{\sin 2x} - 9 - \frac{4 \sec^2 4x}{\tan 4x}$$

$$\Rightarrow \frac{dy}{dx} = y \left[\ln 6 + 2 \cot 2x - 9 - \frac{4 \sec^2 4x}{\tan 4x} \right]$$

- Webaassign wil hē jy moet y
 vervang met $y = \frac{6^x \sin 2x}{e^{9x} \tan 4x}$

Maaf (2) $f(x) = \ln \left[\frac{e^{2x} \tan 2x}{\sec^2(x^2)} \right]$

$$= \ln e^{2x} + \ln \tan 2x - \ln \sec^2(x^2)$$
$$= 2x \ln e + \ln \tan 2x - \ln \sec^2(x^2)$$

Dan is:

$$f'(x) = 2 + \frac{2 \sec^2 2x}{\tan 2x} - \frac{2 \sec(x^2) \times \sec(x^2) \tan(x^2) \times 2x}{\sec^2(x^2)}$$

$$\textcircled{3} \quad y = \frac{\sqrt{x} (x^2-1)^5}{(x+2)(x-4)^3}$$

$$\Rightarrow \ln y = \ln \left[\frac{\sqrt{x} (x^2-1)^5}{(x+2)(x-4)^3} \right]$$

$$\begin{aligned} \Rightarrow \ln y &= \ln \sqrt{x} + \ln (x^2-1)^5 - \ln (x+2) - \ln (x-4)^3 \\ &= \frac{1}{2} \ln x + 5 \ln (x^2-1) - \ln (x+2) - 3 \ln (x-4) \end{aligned}$$

(diff implisiet)

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2x} + \frac{(5)(2x)}{x^2-1} - \frac{1}{x+2} - \frac{3}{x-4}$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{1}{2x} + \frac{10x}{x^2-1} - \frac{1}{x+2} - \frac{3}{x-4} \right]$$

kan y vervang met: $y = \frac{\sqrt{x} (x^2-1)^5}{(x+2)(x-4)^3}$